

## A Matching Model of Unemployment

Continuous Time. Infinite Horizon.

$$\text{Workers} \begin{cases} E = \text{employed} \\ U = \text{unemployed} \end{cases}$$

$$\text{Jobs} \begin{cases} F = \text{filled} \\ V = \text{vacant} \end{cases}$$

Each job can have at most one worker. Then:  $E = F$ .

The labor force,  $L$ , is fixed. Then:  $E(t) + U(t) = L$  at every  $t$ .

We will consider only steady states ( $E$  and  $U$  constant)

The number of jobs is endogenous. Vacancies can be created and eliminated freely.

When the job is vacant the firm is actively engaged in hiring at a fixed cost  $c$  per unit of time.

An employed worker produces output at rate  $y$  per unit of time, and is paid a wage  $w$  per unit of time.

Output ( $y$ ) is exogenous;  $w$  is determined endogenously through *bargaining* between the worker and the firm.

Worker's utility, per unit time:  $u_W = \begin{cases} w & \text{if employed} \\ z & \text{if unemployed} \end{cases}$

We assume  $y > z$ .

Firm's profits from a job, per unit of time:

$$u_J = \begin{cases} y - w & \text{if the job is filled} \\ -c & \text{if the job is vacant} \end{cases}$$

Workers' objective function is the expected present discounted value of utility.

Firms' objective function is the expected present discounted value of profits.

The discount rate,  $r > 0$ , is exogenous and constant (the same for workers and firms).

## Matching

When unemployed workers and vacant jobs meet, new matches (*i.e.*, new filled jobs = new employed workers) are produced at rate  $M$  per unit of time, according to a so-called *matching function* (technology):

$$M = m(U, V)$$

We assume  $m(\bullet)$  is increasing in both arguments, concave, and homogeneous of degree one. Also:  $m(0, V) = m(U, 0) = 0$ .

Example:  $M = AU^\gamma V^{1-\gamma}$  with  $A > 0$  and  $\gamma \in (0, 1)$ .

Define:

$$u \equiv \frac{U}{L} \quad \text{unemp. rate}$$

$$v \equiv \frac{V}{L} \quad \text{vacancy rate}$$

$$m \equiv \frac{M}{L}$$

Remark: we abuse notation and use  $m$  to denote both  $m \equiv \frac{M}{L}$  and the matching function,  $m(\bullet)$ .

Since  $m(\bullet)$  is h-d-one, we can write:

$$m = m(u, v)$$

Define:

$$p \equiv \frac{m(U, V)}{U} = \frac{m(u, v)}{u} = \text{job-finding rate}$$

$$q \equiv \frac{m(U, V)}{V} = \frac{m(u, v)}{v} = \text{job-filling rate}$$

$$\theta \equiv \frac{V}{U} = \frac{v}{u} = \text{labor-market tightness}$$

Since  $m(\bullet)$  is h-d-one, we get:  $p = \frac{m(u, v)}{u} = m(1, \frac{v}{u}) = m(1, \theta)$  and  $q = \frac{m(u, v)}{v} = m(\frac{u}{v}, 1) = m(\frac{1}{\theta}, 1)$ . Now we can write (abusing notation):

$$p = p(\theta) \equiv m(1, \theta)$$

$$q = q(\theta) \equiv m(\frac{1}{\theta}, 1)$$

Notice that  $p' > 0$  and  $q' < 0$ . Also, the concavity of  $m(\bullet)$  implies  $p'' \leq 0$ . Finally, notice that  $p(\theta) = \theta q(\theta)$ .

We assume jobs are destroyed at the exogenous rate  $\alpha$  per unit of time. Hence, during a time interval of length  $h$  we have:  $E(t+h) - E(t) = Mh - \alpha h E$ . Dividing by  $h$  and letting  $h \rightarrow 0$  we get:  $\dot{E} = M - \alpha E$ . In steady state  $\dot{E} = 0$ , then  $M = \alpha E$ . Then:

$$m(U, V) = \alpha E$$

Dividing by  $L$ , using h-d-one, and letting  $e \equiv \frac{E}{L}$  (employment rate) we get:

$$m(u, v) = \alpha e$$

Hence, in s-s, job creation = job destruction.

From  $E + U = L$  we get:

$$e + u = 1$$

From  $m(U, V) = \alpha E$  and  $p \equiv \frac{m(U, V)}{U}$  we can find the steady-state unemployment rate:

$$pU = \alpha E$$

$$pu = \alpha e \quad (\text{dividing by } L)$$

$$pu = \alpha(1 - u)$$

$$(p + \alpha)u = \alpha \Rightarrow$$

$$u = \frac{\alpha}{\alpha + p}$$

Recalling that  $p$  is a function of  $\theta$  we can write:

$$u(\theta) = \frac{\alpha}{\alpha + p(\theta)}$$

## Stationary Value Functions

For a period of length  $h$ , the value of employment is:

$$V_E(h) = wh + \frac{1}{1 + rh} [\alpha h V_U(h) + (1 - \alpha h) V_E(h)]$$

The expression above gives:

$$V_E(h)(1 - \frac{1 - \alpha h}{1 + rh}) = wh + \frac{\alpha h}{1 + rh} V_U(h)$$

$$V_E(h) \frac{(r + \alpha)h}{1 + rh} = wh + \frac{\alpha h}{1 + rh} V_U(h) \Rightarrow$$

$$V_E(h) = \frac{1 + rh}{\alpha + r} w + \frac{\alpha}{\alpha + r} V_U(h)$$

Continuous time:  $h \rightarrow 0$

$$V_E = \frac{w}{\alpha + r} + \frac{\alpha}{\alpha + r} V_U$$

where  $V_E \equiv V_E(0)$  and  $V_U \equiv V_U(0)$ .

The expression above can be rewritten as follows:

$$rV_E = w + \alpha(V_U - V_E)$$

The value of unemployment must satisfy (writing  $p$  instead of  $p(\theta)$ , for simplicity):

$$V_U(h) = zh + \frac{1}{1 + rh} [phV_E(h) + (1 - ph)V_U(h)]$$

The expression above gives:

$$V_U(h)(1 - \frac{1-ph}{1+rh}) = zh + \frac{ph}{1+rh} V_E(h)$$

$$V_U(h) \frac{(r+p)h}{1+rh} = zh + \frac{ph}{1+rh} V_E(h) \Rightarrow$$

$$V_U(h) = \frac{1 + rh}{p + r} z + \frac{p}{p + r} V_E(h)$$

Continuous time:  $h \rightarrow 0$

$$V_U = \frac{z}{p + r} + \frac{p}{p + r} V_E$$

The expression above can be rewritten as follows:

$$rV_U = z + p(V_E - V_U)$$



The value of a filled job must satisfy:

$$V_F(h) = (y - w)h + \frac{1}{1 + rh} [\alpha h V_V(h) + (1 - \alpha h) V_F(h)]$$

The expression above gives:

$$V_F(h) \left(1 - \frac{1 - \alpha h}{1 + rh}\right) = (y - w)h + \frac{\alpha h}{1 + rh} V_V(h)$$

$$V_F(h) \frac{(r + \alpha)h}{1 + rh} = (y - w)h + \frac{\alpha h}{1 + rh} V_V(h)$$

Then:

$$V_F(h) = \frac{1 + rh}{\alpha + r} (y - w) + \frac{\alpha}{\alpha + r} V_V(h)$$

Continuous time:  $h \rightarrow 0$

$$V_F = \frac{y - w}{\alpha + r} + \frac{\alpha}{\alpha + r} V_V$$

The expression above can be rewritten as follows:

$$rV_F = (y - w) + \alpha(V_V - V_F)$$

The value of a vacant job must satisfy (writing  $q$  instead of  $q(\theta)$ , for simplicity):

$$V_V(h) = -ch + \frac{1}{1+rh}[qhV_F(h) + (1-qh)V_V(h)]$$

The expression above gives:

$$V_V(h)(1 - \frac{1-qh}{1+rh}) = -ch + \frac{qh}{1+rh}V_F(h)$$

$$V_V(h)\frac{(r+q)h}{1+rh} = -ch + \frac{qh}{1+rh}V_F(h)$$

Then:

$$V_V(h) = -\frac{1+rh}{q+r}c + \frac{q}{q+r}V_F(h)$$

Continuous time:  $h \rightarrow 0$

$$V_V = \frac{-c}{q+r} + \frac{q}{q+r}V_F$$

The expression above can be rewritten as follows:

$$rV_V = -c + q(V_F - V_V)$$

## Wage Determination (Nash Bargaining)

Once they meet, a firm and a worker bargain over the wage,  $w_i$  ( $i$  indexes this particular firm-worker pair).

The joint payoff from disagreement (no match) is:

$$\mathcal{J}_d = V_U + V_V$$

Remark. The values of  $V_U$  and  $V_V$  do not depend on  $w_i$ , the wage negotiated by this particular firm-worker pair. They are determined by conditions in the rest of the market (not influenced by a particular firm or worker).

The joint payoff from agreement (a match is formed) is:

$$\begin{aligned}\mathcal{J}_a &= V_E(w_i) + V_F(w_i) \\ &= \frac{w_i}{\alpha + r} + \frac{\alpha}{\alpha + r} V_U + \frac{y - w_i}{\alpha + r} + \frac{\alpha}{\alpha + r} V_V \\ &= \frac{y + \alpha(V_U + V_V)}{\alpha + r}\end{aligned}$$

Remark. In our case, the joint payoff from agreement is independent from the negotiated wage (transferable utility).

Surplus from agreement:

$$\begin{aligned} S &= \mathcal{J}_a - \mathcal{J}_d \\ &= \frac{y + \alpha(V_U + V_V)}{\alpha + r} - (V_U + V_V) \\ &= \frac{y - r(V_U + V_V)}{\alpha + r} \end{aligned}$$

Remark 1. Notice that  $S = V_E + V_F - (V_U + V_V) = (V_E - V_U) + (V_F - V_V)$ .

Remark 2.  $S > 0$  requires  $y > r(V_U + V_V)$ . If  $S < 0$  players won't agree.

Nash's Bargaining Solution (NBS):

$$V_E = V_U + \beta S$$

$$V_F = V_V + (1 - \beta)S$$

where  $\beta \in (0, 1)$  is a parameter that can be interpreted as the bargaining power of the worker.

From the expressions above we get:

$$\beta(V_F - V_V) = (1 - \beta)(V_E - V_U)$$

Remark 1. Writing  $V_E(w_i)$  and  $V_F(w_i)$  makes clear that, for given values of  $V_V$  and  $V_U$ , the expression above determines the negotiated wage.

Remark 2. The NBS can be obtained from:

$$\max_{w_i} [V_E(w_i) - V_U]^\beta [V_F(w_i) - V_V]^{1-\beta}$$

Recall that  $V'_E(w_i) = -V'_F(w_i)$ .

To complete the model, recall that vacancies can be created and destroyed costlessly. Hence, in equilibrium:

$$V_V = 0$$

Finally, recall that in a symmetric equilibrium we have

$$w_i = w$$

Summarizing:

$$m = m(u, v)$$

$$m = \alpha e$$

$$\theta = \frac{v}{u}$$

$$u = \frac{\alpha}{\alpha + p(\theta)}$$

$$rV_E = w + \alpha(V_U - V_E)$$

$$rV_U = z + p(\theta)(V_E - V_U)$$

$$rV_F = y - w + \alpha(V_V - V_F)$$

$$rV_V = -c + q(\theta)(V_F - V_V)$$

$$\beta(V_F - V_V) = (1 - \beta)(V_E - V_U)$$

$$V_V = 0$$

Remark. 10 eq. in 10 var.:  $m, u, v, e, \theta, w, V_E, V_U, V_F, V_V$ .

## Solving the Model

We show how to find the equilibrium values of  $w$ ,  $\theta$ ,  $u$  and  $v$ . The equilibrium levels of the rest of the variables follow easily from the equations summarized above.

We start by using the optimization conditions of the firm to find a relationship between  $w$  and  $\theta$ .

From  $rV_F = y - w + \alpha(V_V - V_F)$  we get:

$$V_F - V_V = \frac{(y - rV_V) - w}{r + \alpha}$$

Notice that  $V_F - V_V > 0$  requires  $w < y - rV_V$ . Hence,  $y - rV_V$  is like a reservation wage for the firm.

Substituting  $V_V = 0$  into the expression above we get:

$$V_F = \frac{y - w}{r + \alpha}$$

Hence, the reservation wage for the firm is equal to  $y$  (i.e., the firm won't fill a job if it has to pay  $w > y$ ).

Substituting  $V_V = 0$  into  $rV_V = -c + q(\theta)(V_F - V_V)$  we get:

$$V_F = \frac{c}{q(\theta)}$$

Combining the two expressions above we get:  $\frac{y-w}{r+\alpha} = \frac{c}{q(\theta)}$ . Then:

$$y - w = \frac{r + \alpha}{q(\theta)}c$$

Notice that the r-h-s is positive whenever  $c > 0$ . Hence, the l-h-s implies  $y > w$  (i.e., the wage rate is lower than the firm's reservation wage).

The expression above gives:

$$w = y - \frac{r + \alpha}{q(\theta)}c$$

This is a negative relationship between  $w$  and  $\theta$  (recall  $q$  is decreasing in  $\theta$ ). We call this relationship the *job creation condition* (JC).



Now we combine the opt. conditions of a worker with the NBS to find another relationship between  $w$  and  $\theta$ .

From  $rV_E = w + \alpha(V_U - V_E)$  we get:

$$V_E - V_U = \frac{w - rV_U}{r + \alpha}$$

Remark. We can think of  $rV_U$  as the reservation wage of the worker. Then,  $V_E > V_U$  if and only if  $w$  is higher than the worker's reservation wage (otherwise, the worker will reject the agreement).

From the bargaining condition we know:  $V_E - V_U = \frac{\beta}{1-\beta}(V_F - V_V)$ . But we also know  $V_F = \frac{y-w}{r+\alpha}$  and  $V_V = 0$ . Then:

$$V_E - V_U = \frac{\beta}{1-\beta} \frac{y-w}{r+\alpha}$$

Since  $y > w$ , this expression shows that  $V_E - V_U > 0$ , and then  $w > rV_U$ . Therefore:  $rV_U < w < y$ .

Equating the previous two expressions we get  $\frac{w-rV_U}{r+\alpha} = \frac{\beta}{1-\beta} \frac{y-w}{r+\alpha}$ . Solving for  $w$  we obtain:

$$w = \beta y + (1 - \beta)rV_U$$

Notice that  $w$  is a weighted average between the two reservation wages, and the weights depend on the bargaining powers.

We can eliminate  $rV_U$  from the previous expression by writing it as a function of  $\theta$ . First use  $rV_U = z + p(\theta)(V_E - V_U)$  and the bargaining condition to get:  $rV_U = z + p(\theta)\frac{\beta}{1-\beta}(V_F - V_V)$ . Now recall  $V_F = \frac{c}{q(\theta)}$  and  $V_V = 0$  to get:  $rV_U = z + p(\theta)\frac{\beta}{1-\beta}\frac{c}{q(\theta)}$ . Using  $p(\theta) = \theta q(\theta)$  we get:  $rV_U = z + \theta q(\theta)\frac{\beta}{1-\beta}\frac{c}{q(\theta)} \Rightarrow$

$$rV_U = z + \frac{\beta c}{1 - \beta} \theta$$

Substituting the expression above into the previous one,  $w = \beta y + (1 - \beta)rV_U$ , we get:

$$w = \beta y + (1 - \beta)z + \beta c\theta$$

The expression above gives a positive (linear) relation between labor-market tightness and the equilibrium wage. We call this relation the **wage curve (WC)**.

Combining the job creation condition with the wage curve we can find the equilibrium values of the wage rate,  $w^*$ , and labor-market tightness,  $\theta^*$ .

Equating JC and WC we get:

$$y - \frac{r+\alpha}{q(\theta)}c = \beta y + (1 - \beta)z + \beta c\theta$$

$$(1 - \beta)y = (1 - \beta)z + \beta c\theta + \frac{r+\alpha}{q(\theta)}c$$

$$(1 - \beta)(y - z) = c \left[ \beta\theta + \frac{r+\alpha}{q(\theta)} \right]$$

Then:

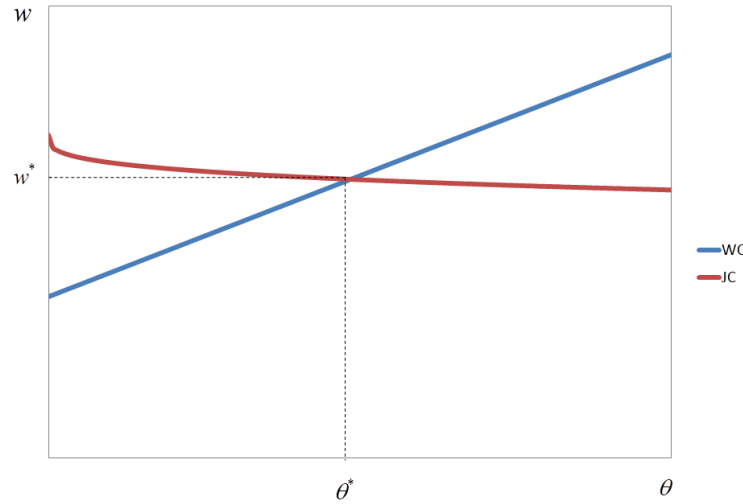
$$\frac{r + \alpha}{q(\theta)} + \beta\theta = (1 - \beta)\frac{y - z}{c}$$

The expression above defines the equilibrium value of labor-market tightness,  $\theta^*$ . Typically, the solution will have to be found numerically.

Then we can go back to WC to get the equilibrium wage:

$$w^* = z + \beta(y - z) + \beta c\theta^*$$

We can represent the determination of  $\theta^*$  and  $w^*$  graphically, by plotting the JC and WC together.



$$\text{JC: } w = y - \frac{r+\alpha}{q(\theta)}c \quad \text{WC: } w = \beta y + (1 - \beta)z + \beta c\theta$$

### Existence

Under standard assumptions on the matching function,  $q(\theta) \rightarrow \infty$  as  $\theta \rightarrow 0$ . Hence, JC will cut the vertical axis at  $y$ . On the other hand, WC cuts the vertical axis at  $\beta y + (1 - \beta)z$ . Hence, if  $y > z$ , the intercept of JC will be above the intercept of WC. Since JC is continuous and decreasing, and WC is continuous and increasing, they will cross at a unique positive value for  $\theta$ . Hence, a unique equilibrium value  $\theta^*$  will exist.

Once we've found  $\theta^*$  we can obtain the equilibrium unemployment rate from:

$$u^* = \frac{\alpha}{\alpha + p(\theta^*)}$$

Recalling that  $\theta = \frac{v}{u}$ , we can obtain the equilibrium vacancy rate from:

$$v^* = \theta^* u^*$$

We can represent the determination of  $u^*$  and  $v^*$  graphically.

Combining  $u = \frac{\alpha}{\alpha + p(\theta)}$  with  $\theta = \frac{v}{u}$  we get:

$$u = \frac{\alpha}{\alpha + p(\frac{v}{u})}$$

The expression above implicitly defines  $v$  in terms of  $u$ . This relation is usually called the *Beveridge Curve* (BC).

Recall that  $p = \frac{m(u,v)}{u}$ . Then, we can rewrite the expression above as follows:

$$\alpha u + m(u, v) = \alpha$$

It is now easy to show that BC has negative slope (recall that  $m(\bullet)$  is increasing in both arguments). Moreover, the concavity of  $m(\bullet)$  implies that BC is convex.

Now we find another relation between  $v$  and  $u$ .

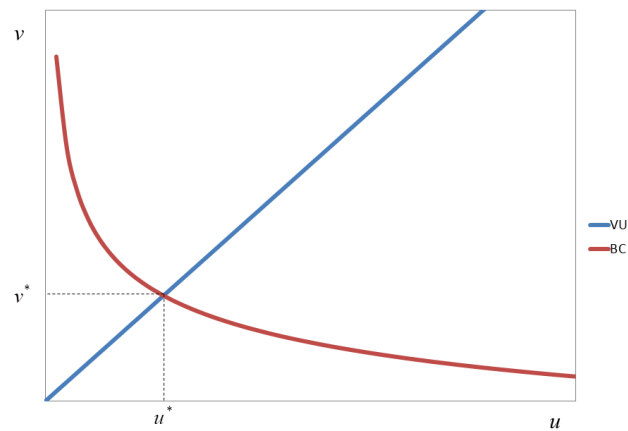
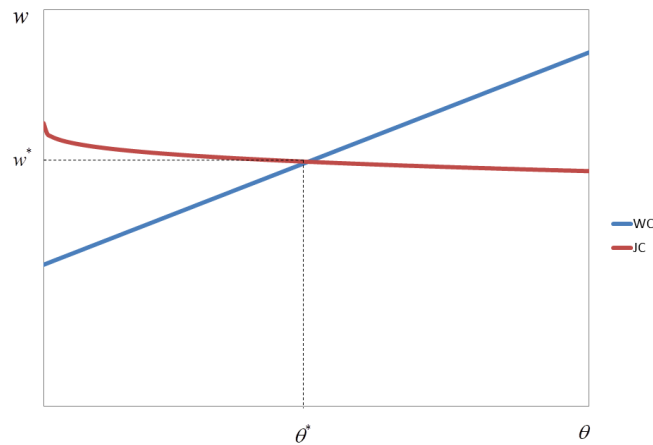
Since  $\theta \equiv \frac{v}{u}$ , we know the equilibrium levels of the vacancy rate and the unemployment rate must satisfy:

$$v = \theta^* u$$

We call the expression above the VU curve (recall  $\theta^*$  has already been determined).

Now we can combine the BC and VU curves in the same plot in order to represent the determination of  $u^*$  and  $v^*$ .

Hence, we can characterize the full equilibrium using the following two plots:

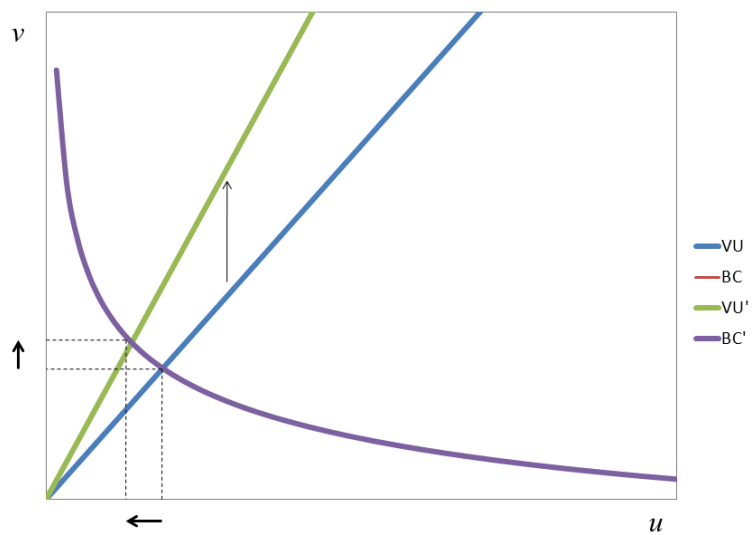
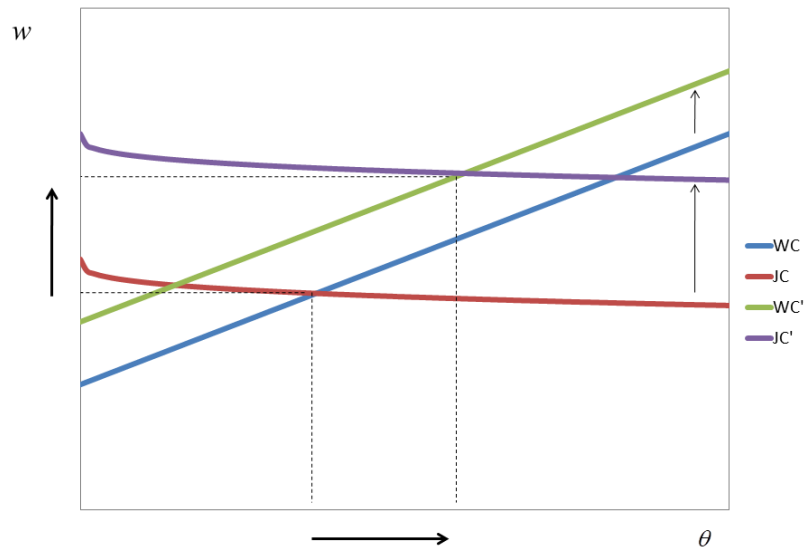


The equilibrium levels of the other variables can be recovered easily from the rest of the equations of the model. For example,  $m^* = m(u^*, v^*)$ ,  $p^* = p(\theta^*)$ ,  $q^* = q(\theta^*)$ , and so on.

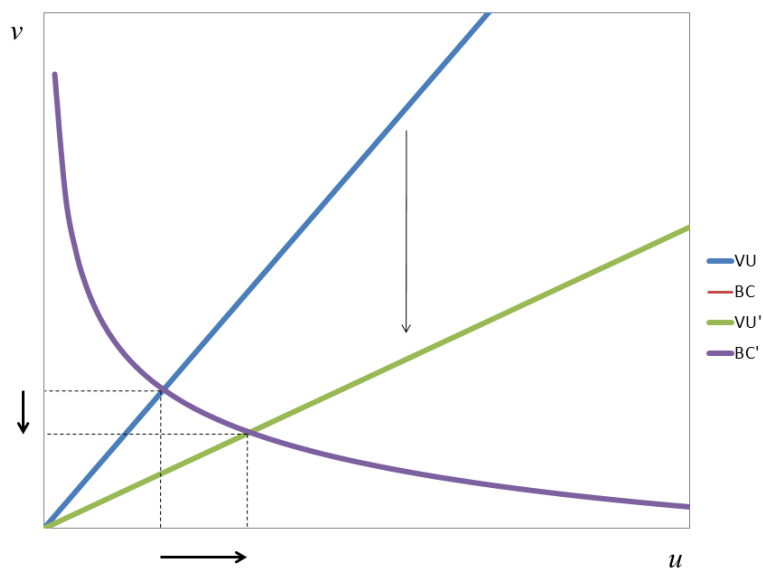
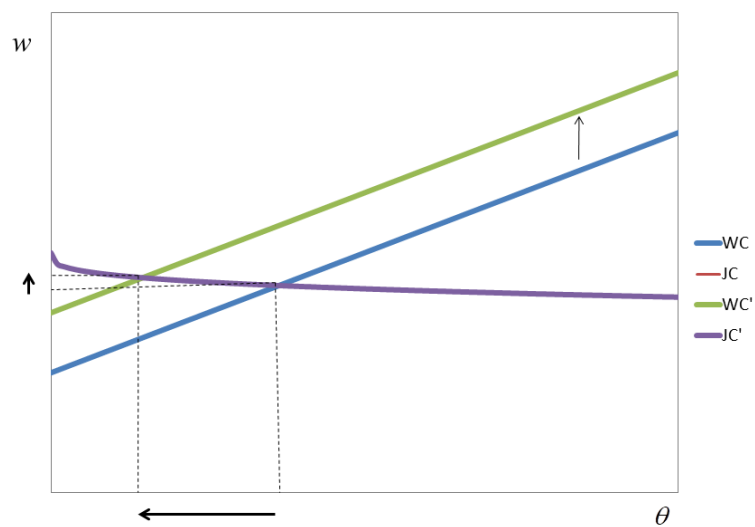


# Comparative Statics

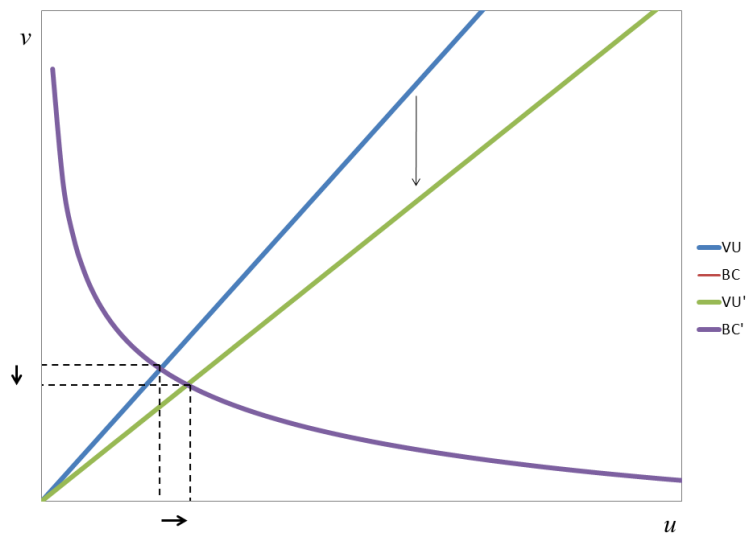
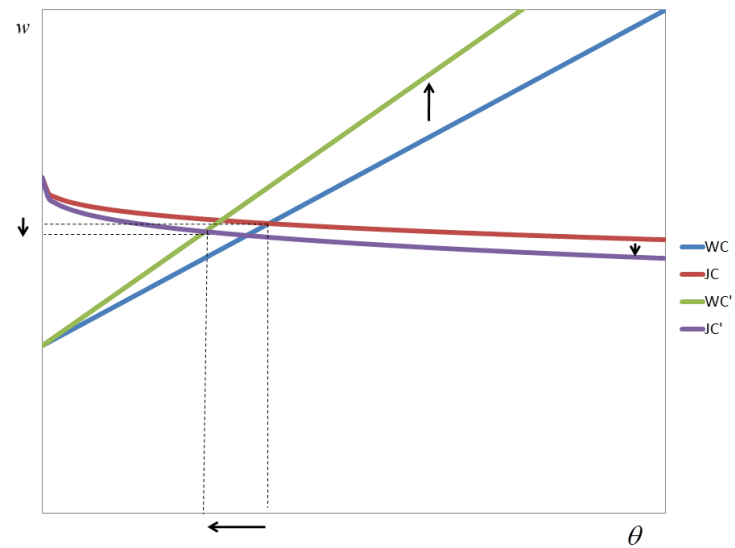
Increase in  $y$



# Increase in $z$

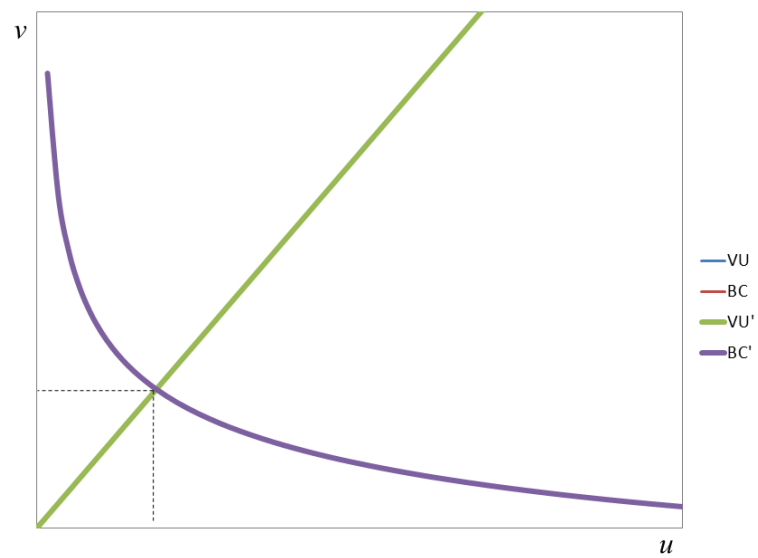
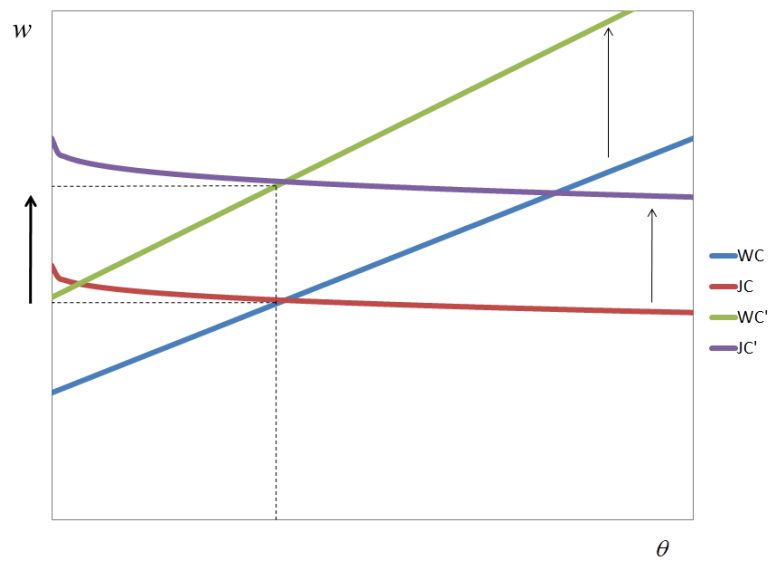


Increase in  $c$

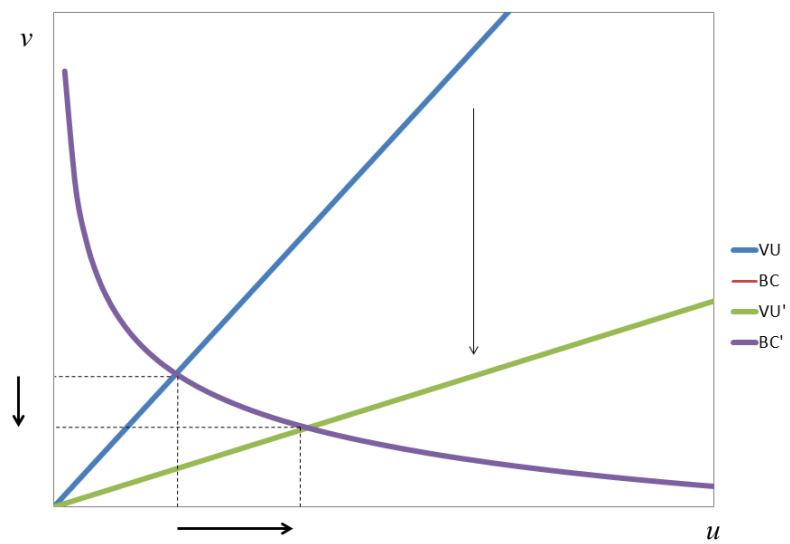
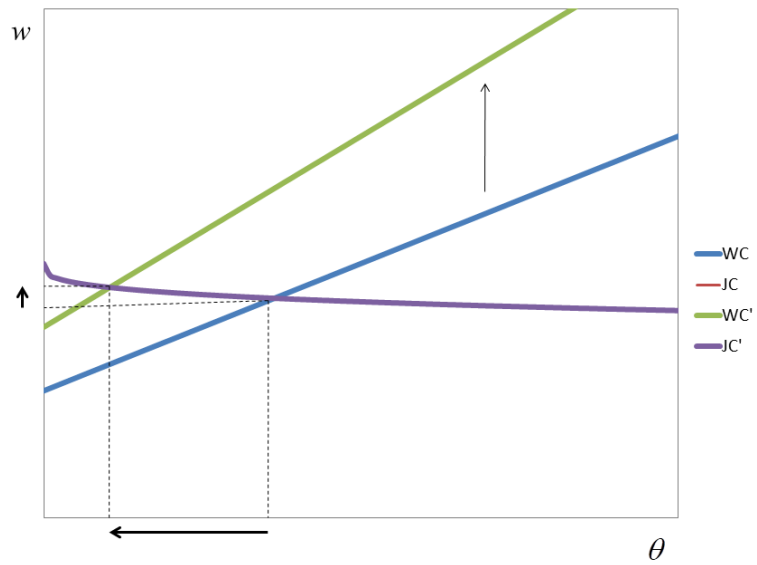


The effect on  $w$  is ambiguous.

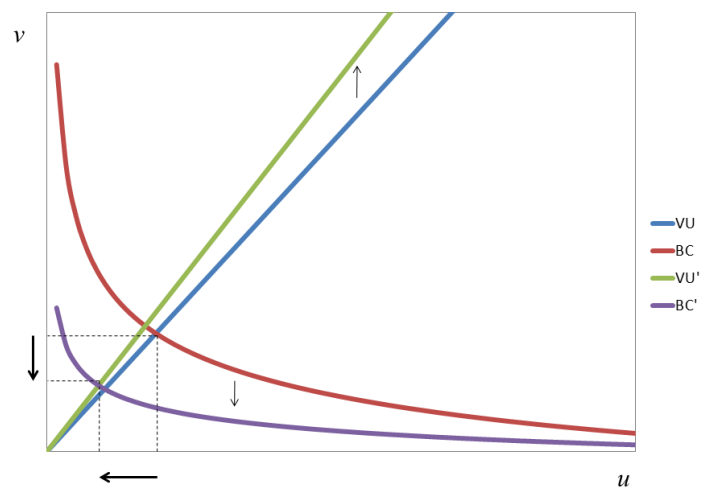
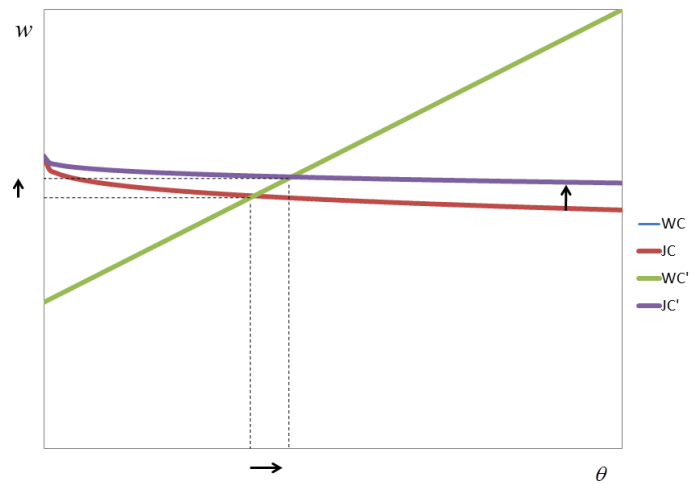
## Proportional increase in $y$ , $c$ , and $z$



## Increase in $\beta$

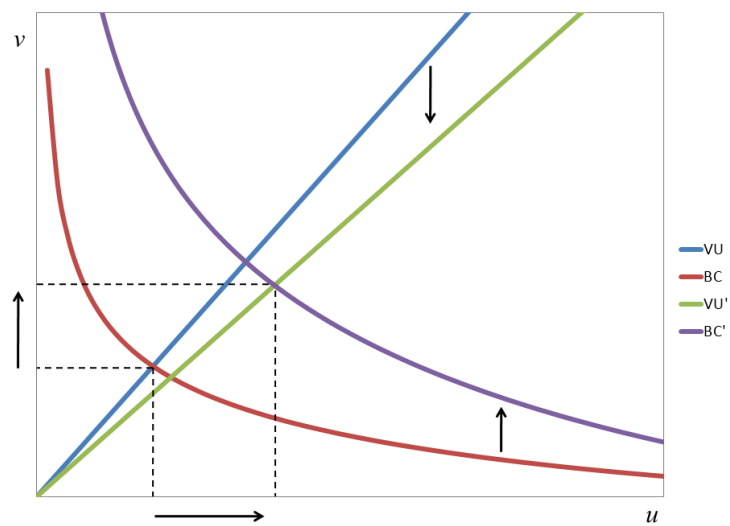
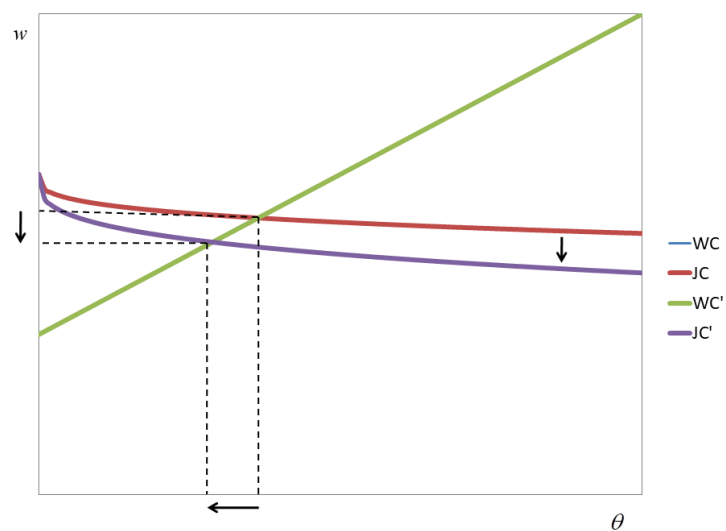


Increase in  $A$  (matching efficiency)



The effect on  $v$  is ambiguous.

Increase in  $\alpha$



The effect on  $v$  is ambiguous.

### Matching and Efficiency

Consider the matching model studied in class. Let say that two paths  $\theta(\bullet)$  and  $u(\bullet)$  are efficient if they solve the following problem:

$$\begin{aligned} \max \quad & \int_0^\infty e^{-rt} [(1-u)y + uz - c\theta u] dt \\ \text{s.t.} \quad & \dot{u} = \alpha(1-u) - p(\theta)u \\ & \theta \in \mathbb{R}_{++} \\ & u(0) = u_0 \\ & \lim_{t \rightarrow \infty} u(t) \geq 0 \end{aligned}$$

where  $r > 0$ ,  $y > 0$ ,  $z > 0$ ,  $c > 0$ ,  $\alpha > 0$ , and  $u_0 \in [0, 1]$  is given.

Remark. For simplicity, we have dropped the dependence of  $u$  and  $\theta$  on  $t$ .

The objective functional is the present discounted value of social output/utility: a fraction  $e = 1 - u$  of employed agents produce  $y$  units of output/utility, while a fraction  $u$  of unemployed agents get  $z$  units of utility/output; in addition, each vacant job  $v = \theta u$  costs  $c$  units of output/utility. The planner is subject to the same matching frictions faced by private agents. Hence, the change in the unemployment rate,  $\dot{u}$ , is equal to the difference between the new unemployed agents generated by the process of job destruction,  $\alpha(1-u)$ , and the new employed agents generated by the matching process,  $p(\theta)u$ . In principle, we should include restrictions of the form  $u \in [0, 1]$  and  $\theta \in \mathbb{R}_+ \forall t$ , but we'll concentrate on interior solutions, where  $u \in (0, 1)$  and  $\theta \in \mathbb{R}_{++}$ . It's clear from the statement of the problem that  $u$  is the state variable and  $\theta$  is the control variable.

We can use optimal control techniques to obtain necessary conditions for a maximum. The current-value Hamiltonian is:

$$H(t, u, \theta, \lambda) = (1-u)y + uz - c\theta u + \lambda[\alpha(1-u) - p(\theta)u]$$

where  $\lambda$  is the co-state variable.

The Maximum Principle establishes that the control variable must maximize the Hamiltonian, and that the co-state variable must satisfy  $\dot{\lambda} - r\lambda = -H_u$ .

A necessary condition for the maximization of  $H$  w.r.t.  $\theta$  is  $H_\theta = 0$ . Then:

$$-cu - \lambda p'(\theta)u = 0 \quad (1)$$

From  $\dot{\lambda} - r\lambda = -H_u$  we get:  $\dot{\lambda} - r\lambda = -[-y + z - c\theta - \lambda\alpha - \lambda p(\theta)]$ . Then:

$$\dot{\lambda} = (y - z + c\theta) + \lambda[r + \alpha + p(\theta)] \quad (2)$$



From (1) we get:  $\lambda p'(\theta)u = -cu \Rightarrow \lambda p'(\theta) = -c \Rightarrow$

$$\lambda = \frac{-c}{p'(\theta)} \quad (3)$$

We know that  $p(\theta) = \theta q(\theta)$ . Then:  $p'(\theta) = q(\theta) + \theta q'(\theta)$ . Then:  $p'(\theta) = q(\theta)[1 + \frac{\theta}{q(\theta)}q'(\theta)] \Rightarrow p'(\theta) = q(\theta)[1 - \frac{-\theta}{q(\theta)}q'(\theta)] \Rightarrow$

$$p'(\theta) = q(\theta)[1 - \eta(\theta)] \quad (4)$$

where

$$\eta(\theta) \equiv \frac{-\theta}{q(\theta)}q'(\theta)$$

Substituting (4) into (3) we get:

$$\lambda = \frac{-c}{q(\theta)[1 - \eta(\theta)]} \quad (5)$$

We know that  $\theta$  is constant in steady state. Hence, (5) implies that  $\lambda$  is constant in steady state, too. Therefore:

$$\dot{\lambda} = 0 \quad (6)$$

Substituting (5) and (6) into (2) gives:

$$\begin{aligned} \dot{\lambda} &= (y - z + c\theta) + \lambda[r + \alpha + p(\theta)] \\ 0 &= (y - z + c\theta) + \frac{-c}{q(\theta)[1 - \eta(\theta)]}[r + \alpha + p(\theta)] \\ 0 &= y - z + c \left[ \theta - \frac{r + \alpha + p(\theta)}{q(\theta)[1 - \eta(\theta)]} \right] \\ 0 &= \frac{y - z}{c} + \theta - \frac{r + \alpha + p(\theta)}{q(\theta)[1 - \eta(\theta)]} \quad \text{dividing by } c \\ 0 &= \frac{y - z}{c} + \theta - \frac{r + \alpha + \theta q(\theta)}{q(\theta)[1 - \eta(\theta)]} \quad \text{using } p(\theta) = \theta q(\theta) \\ \frac{r + \alpha + \theta q(\theta)}{q(\theta)[1 - \eta(\theta)]} - \theta &= \frac{y - z}{c} \\ \frac{r + \alpha + \theta q(\theta) - \theta q(\theta)[1 - \eta(\theta)]}{q(\theta)[1 - \eta(\theta)]} &= \frac{y - z}{c} \\ \frac{r + \alpha + \theta q(\theta) - \theta q(\theta) + \theta q(\theta)\eta(\theta)}{q(\theta)[1 - \eta(\theta)]} &= \frac{y - z}{c} \\ \frac{r + \alpha + \theta q(\theta)\eta(\theta)}{q(\theta)[1 - \eta(\theta)]} &= \frac{y - z}{c} \\ \frac{r + \alpha + \theta q(\theta)\eta(\theta)}{q(\theta)} &= [1 - \eta(\theta)] \frac{y - z}{c} \\ \frac{r + \alpha}{q(\theta)} + \frac{\theta q(\theta)\eta(\theta)}{q(\theta)} &= [1 - \eta(\theta)] \frac{y - z}{c} \end{aligned}$$

Then:

$$\frac{r + \alpha}{q(\theta)} + \eta(\theta)\theta = [1 - \eta(\theta)] \frac{y - z}{c} \quad (7)$$

But we know from our lecture notes that, in steady-state, the equilibrium value of  $\theta$  must satisfy:

$$\frac{r + \alpha}{q(\theta)} + \beta\theta = (1 - \beta) \frac{y - z}{c} \quad (8)$$

Comparing (7) and (8) we conclude that the steady-state equilibrium will be efficient whenever:

$$\beta = \eta(\theta) \quad (9)$$

The expression above is usually called the *Hosios condition*.<sup>1</sup>

*Remarks.*

Notice that  $\eta(\theta)$ , which is the elasticity of  $q(\theta)$  with respect to  $\theta$ , is also the elasticity of the matching function with respect to unemployment. To see this, recall that  $q(\theta) \equiv m(\frac{1}{\theta}, 1) = \frac{m(u, v)}{v}$ . Then:  $\eta(\theta) \equiv -\frac{\theta}{q(\theta)}q'(\theta) = -\frac{\frac{v}{m(u, v)}m_1(\frac{1}{\theta}, 1)\frac{-1}{\theta^2}}{\frac{v}{m(u, v)}} = \frac{v}{u} \frac{v}{m(u, v)} m_1(\frac{u}{v}, 1) \frac{1}{(\frac{v}{u})^2} = \frac{v}{m(u, v)} m_1(\frac{u}{v}, 1) \frac{1}{\frac{v}{u}} = \frac{u}{m(u, v)} m_1(\frac{u}{v}, 1)$ .

Notice also, that  $1 - \eta(\theta)$  is the elasticity of  $p(\theta)$  with respect to  $\theta$ . To see this, recall that  $p(\theta) = \theta q(\theta)$ . Then:  $\frac{\theta}{p(\theta)}p'(\theta) = \frac{\theta}{\theta q(\theta)}[q'(\theta)\theta + q(\theta)] = \frac{1}{q(\theta)}[q'(\theta)\theta + q(\theta)] = \frac{q'(\theta)\theta}{q(\theta)} + 1 = -\frac{q'(\theta)\theta}{q(\theta)} + 1 = 1 - \eta(\theta)$ .

We know  $p'(\theta) > 0$ , then  $\frac{\theta}{p(\theta)}p'(\theta) = 1 - \eta(\theta) > 0$ . Then  $\eta(\theta) < 1$ . We also know  $\eta(\theta) > 0$ . Hence:  $\eta(\theta) \in (0, 1)$ .

We know unemployment increases with  $\beta$  (the bargaining power of the worker). Hence, when  $\beta > \eta(\theta)$  unemployment is above the efficient level. Conversely, when  $\beta < \eta(\theta)$  unemployment is below the efficient level. This is consistent with the fact that both firms and workers are causing congestion for each other. One more hiring firm makes searching workers better off, but it makes other hiring firms worse off. Analogously, one more searching worker makes hiring firms better off, but it makes other searching workers worse off. Hence, equilibrium unemployment can be above or below the efficient level.

Since vacancies are filled at Poisson rate  $q(\theta)$ , the expected duration of a vacant job is  $\varphi(\theta) \equiv \frac{1}{q(\theta)}$ . Since unemployed workers move into employment at a Poisson rate  $p(\theta)$ , the expected duration of unemployment spells is  $\psi(\theta) \equiv \frac{1}{p(\theta)}$ . To provide some rationale for condition (9), notice that the elasticity of  $\varphi$  with respect to vacancies is  $\varepsilon_{\varphi}(\frac{v}{u}) \equiv \frac{\frac{v}{\varphi(\frac{v}{u})} \frac{\partial \varphi(\frac{v}{u})}{\partial v}}{\frac{v}{\varphi(\frac{v}{u})}} = \frac{\frac{v}{\varphi(\frac{v}{u})} \varphi'(\frac{v}{u}) \frac{1}{u}}{\frac{v}{\varphi(\frac{v}{u})}} = \frac{\varphi'(\frac{v}{u})}{\varphi(\frac{v}{u})} \varphi(\frac{v}{u}) = \frac{\theta}{\varphi(\theta)} \frac{d}{d\theta}(\frac{1}{q(\theta)}) = \theta q(\theta) \frac{-q'(\theta)}{[q(\theta)]^2} = -\frac{\theta}{q(\theta)} q'(\theta) = \eta(\theta)$ . Also, the elasticity of  $\psi$  with respect to unemployment is  $\varepsilon_{\psi}(\frac{v}{u}) \equiv \frac{\frac{u}{\psi(\frac{v}{u})} \frac{\partial \psi(\frac{v}{u})}{\partial u}}{\frac{u}{\psi(\frac{v}{u})}} = \frac{\frac{u}{\psi(\frac{v}{u})} \psi'(\frac{v}{u}) \frac{-v}{u^2}}{\frac{u}{\psi(\frac{v}{u})}} = -\frac{\frac{v}{\psi(\frac{v}{u})} \psi'(\frac{v}{u})}{\frac{u}{\psi(\frac{v}{u})}} = -\frac{\theta}{\psi(\theta)} \psi'(\theta) = -\frac{\theta}{\frac{1}{p(\theta)}} \frac{d}{d\theta}(\frac{1}{p(\theta)}) = -\theta p(\theta) \frac{-p'(\theta)}{[p(\theta)]^2} = \frac{\theta}{p(\theta)} p'(\theta) = 1 - \eta(\theta)$ . Hence, if  $\eta(\theta)$  is high (and then  $1 - \eta(\theta)$  is low), it is an indication that at the margin firms are causing more congestion to other firms than the congestion that workers are causing to other workers. In this case, the social planner "taxes" firms by giving workers a higher share in the wage bargain (a higher  $\beta$ ).

<sup>1</sup>Hosios, Arthur J. (1990). "On the Efficiency of Matching and Related Models of Search and Unemployment," *Review of Economic Studies*, Vol. 57, No. 2 (April), pp. 279-298.